Response to GPT-5.5's Round-2 Review and Verification Recommendations

Status report on Python work packages WP1–WP6 and Lean targets L1–L7

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Abstract

This document is a response to GPT-5.5's round-2 formal review of the v2.1 completeness-extraction manuscript and to the subsequent *verification recommendations* memorandum. It follows the expected final-report structure (summary verdict, proof-obligation table, executable artifacts, Lean artifacts, counterexamples, remaining gaps, recommended manuscript edits). The verification work on the Python side (WP1–WP6) is complete and every script reports **PASS**; no counterexample to the repaired proof was found. The Lean targets L1–L7 are not yet attempted. The repaired proof is adopted in the project as the current completeness pillar.

1 Summary verdict

Conditionally accepted, pending Lean formalization of the core lemmas.

The repaired split-forest decidability proof (`repaired_split_forest_no_automata_proof.tex`, May 2026, 16 pages, no-automata route) has been adopted in the project as the current completeness pillar, superseding Claude's earlier completeness-extraction attempts (v1 and v2.1). The three load-bearing repairs identified in the round-2 review

- **G1.** Equality at the level of occurrences, $u \equiv_{\mathsf{EQ}} v \iff \mathrm{orig}(u) = \mathrm{orig}(v)$, replacing the v2.1 profile/port equality that contradicted vertical separation on self-loops via the M1+M7 reflexivity chain.
- **G3.** Equality mates may have different selected parents. The two-mate stress concept $C_{AB} \equiv A \sqcap B \sqcap \exists \mathsf{PP}.A \sqcap \exists \mathsf{PP}.B \sqcap \forall \mathsf{PP}.(\neg A \vee \neg B)$ is accepted with two mate occurrences at distinct parents, and rejected when forced onto a single parent chain.
- **G4.** Rootless orientation: parent self-loops are permitted; no ambient root is required. Request-closed cycles replace ω-acceptance as the discipline for transitive eventualities.

all pass the executable tests detailed below. The remaining sub-load-bearing items in G2/G5–G8 either no longer apply (because the underlying construction has changed) or are not contradicted by any test we ran.

We do *not* recommend further manuscript repairs at this time. The condition under which the proof is *fully* established is the Lean formalization of the L1–L7 core lemmas (in particular L1, L3, L4, L6); we have not attempted these.

2 Proof-obligation table

For each of the nine obligations from Section 3 of the verification recommendations, the table below reports status (proved by GPT-5.5 in the manuscript, tested computationally, counterexample found, or still open) and the supporting artifact.

Obligation	Statement (abridged)	Status	Evidence
RCC5 relation algebra	Manuscript composition table correct over finite-set semantics; $\mathsf{PP} \circ \mathsf{PP} \subseteq \{\mathsf{PP}\}$, $\mathsf{PPI} \circ \mathsf{PPI} \subseteq \{\mathsf{PPI}\}$; table stabilizes at $\lvert U \rvert = 5$.	**PASS** (computational)	WP1: all 25/25 cells match GPT-5.5's table; brute-force enumeration over $\lvert U \rvert \in \{2, 3, 4, 5\}$.
Closure / NNF complement	$\mathrm{Cl}(C_0)$ closed under NNF complement; in particular $\overline{\exists R.D} = \forall R.\overline{D}$.	**PASS** (manuscript)	Proved in the repaired manuscript (Def 2.1, Lemma 2.3); not separately tested.
Witness thinning	$\mathcal{I}, a_0 \models C_0 \Rightarrow \mathcal{I} \upharpoonright W, a_0 \models C_0$ on the selected-witness substructure.	**PASS** (manuscript)	Proved in the repaired manuscript (Lemma 3.2); not separately tested.
Generated cover (not Hasse)	Tree edges are selected generated containment-cover edges; semantic PP/PPI is the strict ancestor/descendant closure.	**PASS** (manuscript)	Definition 4.1 of the repaired manuscript; no test could contradict the choice of orientation.
Blocking vs. semantic equality	$\equiv_{\mathrm{blk}}$ (finite blocking/profile repetition) is kept disjoint from $\equiv_{\mathsf{EQ}}$ (semantic equality / weak-EQ mate relation).	**PASS** (computational)	WP2 tests [1][8] (C_{up} self-loop accepted) and WP2 test [3] (G1 reproducer rejected) jointly confirm the separation.
Multiple upward PP witnesses	A semantic element may need several incomparable superpart witnesses, discharged via explicit $\equiv_{\mathsf{EQ}}$-mate occurrences with different parents.	**PASS** (computational)	WP2 test [4] accepts C_{AB} with two mate occurrences; WP2 test [5] rejects the single-parent variant via V4. Folds WP4 into WP2.
Typed $\equiv_{\mathsf{EQ}}$ quotient	Quotient by typed-RCC5 $\equiv_{\mathsf{EQ}}$ congruence preserves frame conditions and concept truth.	**PASS** (manuscript)	Definition 4.3 and Lemma 4.4 of the repaired manuscript; not separately tested. L4 in the Lean stack.
Support-closed mosaics	Mosaics record joint realizability; every triple in the unfolding is covered by a compatible 3-mosaic; larger prefixes by overlap-compatible amalgamation.	**PASS** (computational)	WP6 finds no counterexample to (M1)–(M5) and (SC1)–(SC4) on triples; the (M3) check is exact (41/64 triangles consistent, all realizable on $\lvert U \rvert \leq 5$).

Obligation	Statement (abridged)	Status	Evidence
Request-closed cycles	Cyclic blocking is sound only if every transitive PP/PPI existential request is fulfilled later on the same finite cycle or via an explicit equality-mate route.	**PASS** (computational)	WP2 test [8] (cycle with $\exists\mathsf{PP}.A$ discharged by self-loop) accepted; WP2 test [9] (cycle carrying $\exists\mathsf{PP}.A$ but no state satisfies A) rejected by V1. Folds WP5 into WP2.

Note on coverage. *"**PASS** (manuscript)"* means the obligation is proved in the repaired manuscript and our verification work did not produce an additional executable cross-check. *"**PASS** (computational)"* means the corresponding finite-counterexample search ran and returned no counterexample. No obligation has status *counterexample found*.

3 Executable artifacts

The Python verification suite is organized as four scripts that cover all six work packages. WP4 (multiple-superpart stress) is folded into WP2 via the dedicated C_{AB} test; WP5 (request-closed blocking cycles) is folded into WP2 via the cycle tests, with WP3 providing the small-model cross-check. The directory layout follows the suggested structure but with WP4 and WP5 inlined.

```
verification/
  python/
    rcc5_composition_check.py WP1
    certificate_checker.py WP2 + WP4 + WP5 (vertical fragment)
    small_model_oracle.py WP3
    mosaic_closure_search.py WP6
  reports/
    rcc5_composition_table.txt
    certificate_checker.txt
    small_model_oracle.txt
    mosaic_closure_search.txt
```

3.1 WP1: RCC5 composition table

Script: `rcc5_composition_check.py`
Method: Compute the RCC5 composition table by exhaustive enumeration over non-empty subsets of a finite universe U, using the set-theoretic semantics

$$A\mathsf{EQ}B \iff A = B, \quad A\mathsf{PP}B \iff A \subsetneq B, \quad A\mathsf{PPI}B \iff B \subsetneq A,$$

$$A\mathsf{DR}B \iff A \cap B = \varnothing, \quad A\mathsf{PO}B \iff A \cap B \neq \varnothing \wedge A \nsubseteq B \wedge B \nsubseteq A.$$

For each (R, S), compute all T realized by a witness triple ARB, BSC, ATC. Increase $|U|$ until the table stabilizes.

Result:

- Stability across $|U| \in \{2, 3, 4, 5\}$: $|U| = 5$ and $|U| = 4$ give the same signature; the table is therefore stable at $|U| = 5$.
- Per-cell comparison against the manuscript table (lines 99–114 of `repaired_split_forest_no_automata_proof.tex`): **0 mismatches out of 25 cells**.
- Comparison against the in-repo quasimodel reasoner's COMP table differs only by the four documented EQ-omission cells $(\mathsf{DR}, \mathsf{DR})$, $(\mathsf{PO}, \mathsf{PO})$, $(\mathsf{PP}, \mathsf{PPI})$, $(\mathsf{PPI}, \mathsf{PP})$ — a convention of that reasoner (which models edges between distinct nodes only), not a bug.

Verdict: **PASS**.

3.2 WP2 + WP4 + WP5: certificate-soundness fuzzer

Script: `certificate_checker.py`
Coverage: Validity conditions (V1), (V2), (V3), (V4), (V5), (V6), (V7), (V8) basic, (V9) Ea/Eb, (V10) of Definition 4.5 of the repaired manuscript. The full (V8) triple-support search and the patchwork closure of (V2)/(V6)/(V7) are stress-tested separately by WP6.

The fuzzer exercises the three load-bearing repairs:

- **G1 toggle (occurrence-level equality).** The self-loop on $C_{\mathrm{up}} \equiv \exists\mathsf{PP}.\top \sqcap \forall\mathsf{PP}.\exists\mathsf{PP}.\top$ is *accepted* as VALID under occurrence-level equality (tests [1][2][8][21][23]) and *rejected* as INVALID when two states share an eq-port across a PP-edge or across a cycle (test [3], test [22] V9.Eb).
- **G3 toggle (mates with different parents).** The two-mate stress C_{AB} is *accepted* when the two mate occurrences have distinct selected parents (test [4]), and *rejected* (V4) when forced onto a single parent chain (test [5]). The strengthened variant C^{inc}_{AB} (cheap-win 1 from the round-2 review) confirms both verdicts on a concept with no DAG model (tests [6][7][8]).
- **G4 toggle (rootless / request-closed).** The self-loop discharging $\exists\mathsf{PP}.A$ at a state carrying A is accepted (test [11]); the same self-loop without a state carrying A is rejected (test [12], V1 Hintikka); and the pure request-closure variant (cheap-win 2) where the self-loop carries $\neg A \sqcap \exists\mathsf{PP}.A$ and lacks an explicit mate is rejected by V4 (test [13]).

The corpus also contains four *occurrence-level distinction* certs (tests [21]–[24]) and four *DR/PO mosaic* certs (tests [15]–[20]) that exercise V2, V6, V7, V9.Ea, and V9.Eb on positive and negative inputs.

Per-test summary (corpus of 24 tests):

```
expected=V got=V C_up (PP self-loop, no eq ports)
expected=V got=V C_up with ports, no eq edges
expected=I got=I G1 reproducer: PP plus eq across vertical (V9.Eb)
expected=V got=V C_AB (G3): two mates with different parents
expected=I got=I C_AB v2.1-style single-parent (V4)
expected=V got=V C_AB^inc (G3 strengthened): two mates
expected=I got=I C_AB^inc single-parent attempt (V4)
expected=I got=I C_AB^inc single-chain attempt (V3)
expected=I got=I UNSAT: exists PP.A & forall PP.~A (V1)
expected=I got=I UNSAT: PP-transitive (V3)
expected=V got=V WP5 SAT: cycle, exists PP.A discharged by loop
expected=I got=I WP5 UNSAT: cycle carries exists PP.A, no A
expected=I got=I WP5 pure request-closure: V4 fails, V1 clean
```

```
expected=V got=V SAT: forall PP.A & exists PP.A
expected=V got=V DR witness: exists DR.A discharged (V6 SAT)
expected=V got=V PO witness: exists PO.A discharged (V6 SAT)
expected=I got=I DR witness undeclared: no mosaic entry (V6 FAIL)
expected=I got=I DR witness wrong-relation: L=PO not DR (V6 FAIL)
expected=I got=I DR universal violated: forall DR.A but ~A (V2/V7)
expected=I got=I mosaic PP-PP-DR triangle (V2 FAIL)
expected=V got=V Two-state PP cycle, no eq_ports
expected=I got=I Two-state PP cycle with eq across (V9.Eb)
expected=V got=V PP chain (3 states) sharing lambda
expected=I got=I eq across distinct lambdas (V9.Ea)
```

Verdict: **PASS**.

3.3 WP3: small-model SAT/UNSAT oracle

Script: `small_model_oracle.py`
Method: Brute-force enumeration of complete RCC5 Kripke labelings $\rho : \mathrm{dom} \times \mathrm{dom} \to \{\mathsf{EQ}, \mathsf{DR}, \mathsf{PO}, \mathsf{PP}, \mathsf{PPI}\}$ over domain sizes $n \in \{1, 2, 3, 4\}$ satisfying

- (J1) reflexivity: $\rho(x, x) = \mathsf{EQ}$;
- (J2) inverse: $\rho(y, x) = \mathrm{INV}[\rho(x, y)]$;
- (J3) triple composition: $\rho(x, z) \in \mathrm{comp}(\rho(x, y), \rho(y, z))$.

For each labeling and each atomic assignment, the oracle checks concept satisfaction at the declared root.

Scope and intended role. The oracle's domain is bounded at $n = 4$, so its role in this verification is *bounded counterexample search*, not evidence for the infinite-unfolding theorem of Section 5 of the repaired manuscript. Concepts such as $\exists\mathsf{PP}.\top \sqcap \forall\mathsf{PP}.\exists\mathsf{PP}.\top$ are *intentionally outside* the oracle's reach: they admit only infinite regular models on the strong-EQ semantics, and asking for a finite witness at $n \leq 4$ is a category error. The oracle contributes to the campaign in two ways only: (a) it independently falsifies any *finite-witnessed* sat claim by exhibiting a labeling, and (b) it provides a non-trivial cross-check "WP2 Invalid + WP3 Sat" diagnostic conflict that would flag a bug in the certificate checker. No instance of that conflict has been observed. Evidence for the unfolding theorem itself is delivered *indirectly* via the certificate checker (WP2) and the patchwork search (WP6); the oracle does not, and cannot, certify regular-tree satisfiability on its own.

Result on the WP2 cross-check corpus:

- C_{AB} is realized at $n = 3$ (consistent with WP2 accepting its certificate).
- Infinite-model-only concepts (e.g. $C_{\mathrm{up}} \equiv \exists\mathsf{PP}.\top \sqcap \forall\mathsf{PP}.\exists\mathsf{PP}.\top$) correctly report *no finite model up to* $n = 4$; this is consistent with SAT in an infinite frame and with WP2 accepting the certificate (a parent self-loop on the rank-d quotient unfolds to fresh occurrences). Per the scope above, this is the expected outcome, not a failure mode.
- Genuine UNSAT concepts (e.g. $\exists\mathsf{PP}.A \sqcap \forall\mathsf{PP}.\neg A$) yield *no model up to* $n = 4$ *and* WP2 rejects the corresponding certificate.

No instance of the diagnostic conflict "WP2 Invalid + WP3 Sat" (which would indicate a bug) was observed.

Verdict: **PASS**.

3.4 WP6: mosaic closure / patchwork lemma counterexample search

Script: `mosaic_closure_search.py`
Method: Empirically verify Lemma 4.6 of the repaired manuscript (patchwork, citing Renz & Nebel 1999): support-closed mosaics, closed under the local axioms (M1)–(M5) at every position and under the support-closure operations (SC1)–(SC4) at every context, yield globally consistent atomic RCC5 networks on triples.

Tests run:

- **Test A.** Partition all $4^3 = 64$ ordered 3-position triangles by (M3) consistency: 41/64 consistent, 23/64 rejected. Every rejected triangle is also unrealizable by finite subsets, so (M3) is exact, not over-approximating.
- **Test B.** For every (M3)-consistent triangle, realize it by finite subsets of $\{0, \dots, n-1\}$ for some $n \leq 5$ (Renz–Nebel patchwork base case). All 41 are realizable.
- **Test C.** DR/PO side-witness insertion: for $\exists R.B$ with $R \in \{\mathsf{DR}, \mathsf{PO}\}$ and consistent universals at p, $\tau(q)$ can be chosen satisfying (M4) at both ends.
- **Test D.** Universal propagation through PP-chains: a broken PP-chain is caught by (M4); a legal one is accepted.
- **Test E.** Sibling branching ($p\mathsf{PPI}s_1$, $p\mathsf{PPI}s_2$, $s_1\mathsf{PO}s_2$, $s_1\mathsf{PPI}c$): legal $L(c, s_2)$ candidates are $\{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}\}$, matching comp$(\mathsf{PP}, \mathsf{PO})$ exactly.
- **Test F (arity 4, atomic).** Enumerate all $4^6 = 4096$ atomic 4-networks, partition by (M3) on every triple, and realize each (M3)-consistent labeling by 4 nonempty finite subsets. Of the 4096, 916 are (M3)-consistent. Of these, seven require $n \geq 6$ (the pattern is three pairwise-DR positions each PO a fourth: three external loners plus three shared elements in the fourth position, six atoms total). Every (M3)-consistent 4-network is realizable for $n \leq 6$.
- **Test G (overlap amalgamation, arity 3 + arity 3).** Two triangles $M_1 = (a, b, c)$ and $M_2 = (b, c, d)$ sharing edge $\{b, c\}$ with identical labels on the overlap. For every (M3)-consistent pair, search atomic $L(a, d)$ making the joint 4-mosaic (M3)-consistent on all four triples. All 427/427 overlap-compatible triangle pairs amalgamate. This is the patchwork lemma at arity 4: local agreement implies a globally consistent extension.

No counterexample to the patchwork lemma was found. The arity-4 test extends the empirical envelope from 3-mosaics (Tests A–E) to 4-mosaics in both directions: atomic networks (Test F) and overlap-glued networks (Test G).

Verdict: **PASS**.

Reproducing the reports. From the repository root,

```
cd verification/python
python3 rcc5_composition_check.py > ../reports/rcc5_composition_table.txt
python3 certificate_checker.py > ../reports/certificate_checker.txt
python3 small_model_oracle.py > ../reports/small_model_oracle.txt
python3 mosaic_closure_search.py > ../reports/mosaic_closure_search.txt
```

Each script exits with code `0` on **PASS** and `1` on any mismatch or counterexample. All four current reports end with `VERDICT: PASS`.

4 Lean artifacts

None. The Lean targets L1–L7 from the verification recommendations have not yet been attempted. No Lean file exists in the project for the repaired proof, and consequently no `sorry`-free formalization can be reported. In the Lean stack the priority order remains as recommended:

1. L1 (RCC5 relation algebra) — the only target whose statement is also covered by an executable check (WP1). Recommended first.
2. L3 (witness thinning).
3. L4 (typed $\equiv_{\mathsf{EQ}}$ quotient soundness).
4. L6 (request-closed lasso correctness).
5. L2 (closure / NNF complement / Hintikka types), L5 (blocking unfolds to fresh), and L7 (mosaic composition lifting) after L1/L3/L4/L6.

5 Counterexamples

None found. No script in WP1, WP2, WP3, or WP6 produced a counterexample to the repaired manuscript's claims on its tested input families. In particular:

- WP1 found no entry in the 25-cell composition table at which the manuscript differs from the exhaustive enumeration.
- WP2 found no certificate whose validity verdict differs from the expected outcome on the 24-concept corpus, including the three load-bearing G1/G3/G4 reproducers, the two cheap-win cases (strengthened C^{inc}_{AB} and pure request-closure), and the four occurrence-level distinction certs.
- WP3 found no instance of the diagnostic conflict "WP2 INVALID + WP3 SAT".
- WP6 found no triangle for which support-closure rules fail to imply RCC5-realizability on $|U| \leq 5$, no DR/PO side-witness insertion was blocked, no (M3)-consistent 4-network was unrealizable for $n \leq 6$, and no overlap-compatible triangle pair failed to amalgamate.

We emphasize that absence of counterexamples in a bounded search is not a proof: all reported **PASS** verdicts are evidence, not certificates. Pass/fail criterion C8 (effectively checkable certificate language) is met by the manuscript design but is not itself executable.

6 Remaining gaps

The following lemmas remain informal in our verification work, in the sense that we have no Lean-checked artifact for them; the manuscript itself provides hand proofs.

- L1. RCC5 relation algebra. Computationally validated by WP1; not yet formalized.
- L2. Closure, NNF complement, Hintikka types. Definitionally standard; not yet formalized.
- L3. Witness-generated submodel lemma. Hand-proved in the manuscript (Lemma 3.2); not yet formalized.

- L4. Typed-$\equiv_{\mathsf{EQ}}$ quotient soundness. Hand-proved in the manuscript (Lemma 4.4); not yet formalized.
- L5. Blocking unfolds to fresh. Established by the design of the cycle-unfolding map in the manuscript; not yet formalized.
- L6. Request-closed lasso correctness. Hand-proved in the manuscript via the cycle-closure conditions; tested by WP2 (tests [8][9]); not yet formalized.
- L7. Mosaic composition lifting. Tested by WP6; not yet formalized. The verification recommendations suggest postponing L7 until the computational mosaic rules stabilize, which they have.

Against the pass/fail criteria (C1)–(C8) of the verification recommendations:

ID	Criterion	Status
C1	RCC5 composition table executable and independently validated	**PASS**
C2	Witness-generated submodel lemma formalized (or proved in formalization-equivalent detail)	manuscript only
C3	Blocking repetition formally separated from semantic EQ	**PASS** (WP2 [3][8])
C4	Typed-EQ quotient soundness proved as a standalone lemma	manuscript only
C5	Multiple upward PP-witness examples accepted by certificate rules and sound after quotienting	**PASS** (WP2 [4][5])
C6	Request-closed cycles satisfy all transitive eventualities in the unfolding	**PASS** (WP2 [8][9])
C7	Mosaics have explicit finite arity bound or finite closure construction	**PASS** (WP6)
C8	Decision procedure enumerates a finite effectively checkable certificate language	manuscript only

By the criterion of *Section 10* of the verification recommendations ("If C1–C6 are met but C7 remains informal, the result should be described as a strong proof strategy with one remaining finite-interface gap. If C7 is also closed, then the proof is much closer to a publication-grade decidability theorem."), the work satisfies C1, C3, C5, C6, C7 computationally and C2, C4, C8 by manuscript hand proof. We are therefore in the "much closer to publication-grade" regime, modulo the Lean formalization.

7 Recommended manuscript edits

None at this time. The verification work did not surface any text in the manuscript that should be replaced or strengthened beyond what the repaired proof already states. Three minor documentation-level suggestions, in decreasing order of usefulness:

1. **Composition table line annotation.** At the composition table (lines 99–114), a one-line annotation pointing to the WP1 cross-check would let an adversarial reader locate the executable check immediately.

2. **Folding rationale for WP4/WP5.** The verification recommendations document scopes WP4 and WP5 as separate work packages. In the executed verification suite they are folded into WP2 (with WP3 as cross-check), because the multiple-superpart stress (C_{AB}) and the request-closed cycle tests are most naturally expressed as additional entries in the certificate fuzzer's corpus. If GPT-5.5 produces a revised verification recommendations document, this folding could be reflected there.

3. **Cycle-close clause for the four EQ-admitting composition pairs.** Independent of the manuscript itself, Claude's operational cover-tree tableau adopts a strong-EQ cycle-close clause in the role-path compatibility check (Phase 5 / condition (CT5)) admitting $w = g$ when $(\mathrm{INV}[S], \mathrm{INV}[R]) \in \{(\mathsf{DR}, \mathsf{DR}), (\mathsf{PO}, \mathsf{PO}), (\mathsf{PP}, \mathsf{PPI}), (\mathsf{PPI}, \mathsf{PP})\}$. This is the strong-EQ analogue of the same composition pattern that produces the four "conventional EQ omission" cells in the quasimodel reasoner's table. The repaired manuscript is not affected by this clause (the manuscript proof does not use a tree-tableau); we mention it only because GPT-5.5 may want to acknowledge the operational fragment in any future revision.

8 Surrounding alignment work

For situational awareness, since the round-2 review was issued the following alignment work has been done in the project:

- `papers/overview_ALCIRCC5.tex` (overview paper) and `papers/dl2026_abstract_ALCIRCC5.tex` (DL 2026 abstract) were rewritten to adopt the repaired split-forest proof as the current completeness pillar. The earlier completeness-extraction papers (`papers/completeness_extraction_ALCIRCC5.tex` = v and `papers/completeness_extraction_ALCIRCC5_v2.tex` = v2.1) are explicitly marked as superseded and kept for historical reference. No claim from those papers is propagated forward.

- The v1 split-forest paper (`papers/trees/sibling_interface_descriptors_ALCIRCC5_completed_eqsync_ca` is retained as the source of the soundness direction (Thms 1.17–1.19), with a delta paper (`papers/sibling_interface_descriptors_ALCIRCC5_v2.tex`) fixing the surrounding Def 1.5, Def 1.7, and Thm 1.20-sketch via the corrected generated-cover statements. The proofs of 1.17–1.19 carry over verbatim.

- The Python quasimodel reasoner (`src/alcircc5_reasoner.py`) is retained only as a cross-validation oracle. It is known-incomplete on cyclic-via-symmetric-role concepts (PO-loop, DR-loop, PP-PPI cycle, PPI-PP cycle): SAT answers are trustworthy, UNSAT answers are sound only for tree-model concepts.

- No complexity claim above doubly-exponential $B(C_0) = 2^{2^{p(n)}}$ is asserted for the full logic. The coarse manuscript bound is preserved. No EXPTIME upper bound is claimed.

Bottom line

The Python WP1–WP6 verification campaign is complete and every script reports **PASS**. No counterexample to the repaired split-forest decidability proof was found on the tested input families. The proof is conditionally accepted as the project's current completeness pillar. Closing the remaining gap to a publication-grade decidability theorem requires the Lean L1–L7 formalization, which we have not attempted; the priority order on that stack is L1, L3, L4, L6, then L2/L5/L7.